Naman Singh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.69
ITL PUBLIC SCHOOL	Class XII (CBSE)	2024	95.60%
ITL PUBLIC SCHOOL	Class X (CBSE)	2022	96.20%

SKILLS

Hardware & Digital Design: Verilog (RTL design) | FPGA flow: Yosys, NextPNR, Gowin EDA, openFPGALoader

Simulation & Verification: Icarus Verilog, GTKWave, LTspice

Programming: C/C++, Python | UART (custom protocol -- framing, batching)

Tools & Platforms: Git/GitHub, Linux, Windows

WORK EXPERIENCE/RESEARCH PROJECTS

AI Accelerator (NPU) Development (NDA) | *Verilog, FPGA Verification*

- Surveyed **quantization schemes** for NPU compute across accuracy, memory, and compute trade-offs.
- Findings informed the team's choice of quantization scheme for the accelerator's compute datapath.
- Leading **FPGA-based verification** of the accelerator on the **AMD-Xilinx ZCU104.**
- **Benchmarking** planned against edge AI platforms **(NVIDIA Jetson, AMD-Xilinx Kria KV260)** post-verification.

MOSFET Hysteresis Modeling and Simulation | *Python, TCAD*

Research Project under Dr. Shubhadeep Bhattacharjee

- **Implemented a physics-based compact MOSFET** model in **Python**, with core equations extracted from the reference paper.
- **Built a framework for parameter fitting** (differential evolution) and simulation around the model.
- **Validated** against experimental data and plots from the paper; modeled SDG N-FET with **99.6%** accuracy.
- Extended model to **velocity saturation & trap-charge effects**, achieving **>95% accuracy.**

PROJECTS

FPGA-Based K-Means Clustering Accelerator | *Verilog, OSS CAD Suite*

- Architected parallel RTL compute engine (distance cores + comparator) for 3-way classification in **2 clock cycles** (<1 µs).
- **Deployed** and validated the compute engine on a **Gowin Tang Nano 9K FPGA** using Yosys and NextPNR.
- Diagnosed system bottleneck: single-pixel UART handshake capped throughput at **19.4%** of link capacity, not compute.
- Redesigned host-FPGA protocol with **8,192-pixel BRAM batching & framing**, achieving **5.18x speedup**.
- Validated performance at **99.6%** of theoretical UART ceiling; resolved USB-to-UART buffer overflow during testing.

5-Stage Pipelined RISC-V (RV32I) Processor | *Verilog, Icarus Verilog/GTKWave*

- Architected **5-stage pipelined** datapath with dedicated RTL for ALU, control, hazard, and pipeline register units.
- **Hazard resolution:** 2 forwarding paths (MEM/WB-to-EX), stalling for Load-Use hazards & auto-flush on branches.
- **Built and validated** single-cycle and pipelined RTL via Icarus Verilog testbenches and GTKWave waveform analysis.
- Tested **10 core RV32I instructions** across R/I/S/B/J formats to validate control logic & hazard handling.

Drift-Diffusion based MOSCAP Simulator | *C, OpenMP, gnuplot*

- Developed a **transient drift-diffusion 1-D MOSCAP simulator** in C with **variable generation rate**.
- System couples a Poisson solver with the carrier-continuity solver via Gummel iterations for self-consistency.
- **Built the carrier-continuity solver** using Scharfetter-Gummel discretization and Crank-Nicolson time-stepping.
- Parallelized per-bias computation using **OpenMP** across bias points, achieving a **10x** runtime speedup.

RELEVANT COURSES

Digital Design & Architecture: Digital Systems, Digital Systems Lab, Microprocessor & Computer Architecture, Introduction to HDL

Devices & Fabrication: Semiconductor Device Fundamentals, Semiconductor Devices Lab, Introduction to CMOS Processing, Numerical Methods for Device Modeling

Foundations: Circuits & Network Analysis, Analog Electronics, Signals & Systems, Data Structures & Applications

SCHOLASTIC ACHIEVEMENTS

- **Award for Excellence**, Outstanding Academic Performance with Proficiency in Mathematics - Class XII (2023-24)
- **Proficiency Awards** in Physics & Mathematics -- awarded for highest marks in school - Class XI (2022-23)
- **MVPP Scholar** (Govt. of Delhi science/math talent exam), selected among **top 1,000** statewide - Class IX (2020-2021)

POSITIONS OF RESPONSIBILITY

Teaching Assistant - **Engineering Electromagnetics (EE1204)** | *IIT Hyderabad*

- Designed and curated problem sets for a cohort of **120 students**, reinforcing weekly lecture concepts through practice.
- Conducted tutorials and resolved student doubts, supporting exam and assignment evaluation under faculty supervision.

Coordinator - Ideation | *Tinkerer's Lab, IIT Hyderabad*

- Ideated and co-hosted the **Speed Typing Competition** @ **THRUST'26**, IIT Hyderabad's flagship techfest.
- Ideated and hosted two **Game Night** events for TL sponsored by **@Portronics**.
- Managed PR and outreach for **Tinker Tech Camp**, engaging over **600 student participants**.

Assistant coordinator - Public relationships and outreach |*Tinkerer's Lab, IIT Hyderabad*

- Managed **Tinkerer's Lab's Instagram page (2K+ followers)**, creating content for posts for public outreach.

EXTRACURRICULAR

- Member of **NSO Table Tennis** at IIT Hyderabad, participating in training and competitions